

GUIDE K35

Protocol Reference

GUIDE K35 Series exposes standard NMEA 0183 output, RTCM v3.x correction input, and PAIR proprietary commands for embedded-module configuration. This reference is generated from the published Documentation protocol source.

GUIDE K35 Series exposes standard NMEA 0183 output, RTCM v3.x correction input, and PAIR proprietary commands for embedded-module configuration. Use this reference to review the UART-based data interfaces, query module status, adjust GNSS behavior, and configure serial output.

1. Standard data interfaces

GUIDE K35 Series uses the carrier-board UART0 data path for positioning output, module configuration, and RTCM correction-data input. NTRIP transport is typically handled by the host application; the module consumes the RTCM data stream.

NMEA 0183 output

GUIDE K35 Series outputs standard NMEA 0183 positioning sentences through the primary carrier-board UART0 interface. The host application can select sentence types and output intervals with PAIR commands.

RTCM v3.x correction input

GUIDE K35 Series accepts RTCM v3.x correction data for RTK positioning. The host application forwards the correction stream to the module through the carrier-board serial data path.

GUIDE is a surface-mount module rather than an enclosed USB receiver. Route UART0, power, ground, RF, and any required timing or dead-reckoning signals through the carrier board. Review Pin Definitions & Interfaces before finalizing the host connection.

2. PAIR command packet structure

GUIDE K35 Series uses the Airoha AG3335 module platform and exposes its PAIR command set through the carrier-board serial interface. All GUIDE proprietary commands and responses follow the same string format. The maximum length of a command packet is 255 bytes.

```
$PAIR<Command_ID>[, <Data_Field_1>, ..., <Data_Field_N>]*<Checksum><CR><LF>
```

Configuration changes are not automatically stored across power cycles. Send \$PAIR513 after changing persistent settings, then reboot the module or cycle the power supply.

3. PAIR command quick reference

Command	Type	Purpose
System status & control		
PAIR001	Output	Return the PAIR command execution result.
PAIR020	Query	Query firmware release information.
PAIR024	Query	Query the GNSS chipset version.
PAIR004	Set	Restart using available navigation data stored in NVRAM.
PAIR005	Set	Restart without using stored ephemeris data.
PAIR006	Set	Restart without using stored navigation data.
PAIR023	Set	Reboot the GNSS chipset immediately.
PAIR513	Set	Store the current configuration in NVRAM.
PAIR514	Set	Clear NVRAM and restore factory defaults.
GNSS configuration		
PAIR066	Set	Enable or disable satellite constellations.
PAIR067	Query	Query the current constellation settings.
PAIR050	Set	Set the navigation update interval. K35C is specified for 1 Hz; K35E supports up to 10 Hz.
PAIR051	Query	Query the current navigation update interval.
PAIR031	Query	Query the WGS84 ECEF XYZ Cartesian velocity vector.
Interface & formatting		
PAIR864	Set	Configure the UART baud rate.
PAIR865	Query	Query the current UART baud rate.
PAIR062	Set	Configure the output interval for a selected NMEA sentence.
PAIR063	Query	Query the output interval for a selected NMEA sentence.
PAIR098	Set	Configure fractional precision for NMEA position values.
PAIR099	Query	Query the current NMEA decimal precision mode.
PAIR100	Set	Configure the NMEA protocol output mode.
PAIR101	Query	Query the current NMEA protocol output mode.

4. Detailed PAIR command dictionary

\$PAIR001 - Command acknowledgement

Return format

\$PAIR001,<Command_ID>,<Result>*<Checksum><CR><LF>

Return parameters

Field	Description
Command_ID	The three-digit command ID being acknowledged.
Result	0 : Command successfully sent. 1 : Command is processing; wait for the result. 2 : Command sending failed. 3 : Command ID is not supported. 4 : Command parameter error, including out-of-range, missing, or checksum errors. 5 : MNL service is busy; retry shortly.

Example

// Host send: cold start \$PAIR006*3C // Receiver response: success acknowledgement \$PAIR001,006,0*3D

\$PAIR020 - Get version

Format

\$PAIR020*38<CR><LF>

Return format

The module returns a standard acknowledgement packet followed by detailed version information.

\$PAIR020,<Project_Version>,<Frequency>,<SW_package>,<Service_version>,<Service_build_time>,<DSP_L1_rom_version>

Return parameters

Field	Description
Project_Version	Format: <Project_board>_<SDK_version>_<SDK_Build_time> Example: AG3335A_V1.0.0_YYYYMMDD The returned project version may vary by GUIDE module variant and firmware release.
Frequency	S : Single frequency D : Dual frequency (L1/L5)
SW_package	N : Normal W : RAW R : RTK I : NavIC F : Fusion
Service_version	MNL service version, seven characters.
Subsystem versions	DSP L1 and L5, Kernel, KF, RTK, and Fusion version values. Each version is followed by its build time in yyMMDDhhmm format.

Example

// Host send \$PAIR020*38 // Receiver response: placeholder values shown for field structure \$PAIR001,020,0*39 \$

\$PAIR024 - Get chip version

Format

\$PAIR024*3C<CR><LF>

Return format

\$PAIR024,<CHIP>*<Checksum><CR><LF>

Return parameters

Field	Description
CHIP	Chipset version, such as A , M , or S .

Example

```
// Host send $PAIR024*3C // Receiver response $PAIR001,024,0*3D $PAIR024,M*5D
```

\$PAIR004 - Hot start**Format**

```
$PAIR004*3E<CR><LF>
```

Return format

```
$PAIR001,004,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send $PAIR004*3E // Receiver response $PAIR001,004,0*3F
```

\$PAIR005 - Warm start**Format**

```
$PAIR005*3F<CR><LF>
```

Return format

```
$PAIR001,005,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send $PAIR005*3F // Receiver response $PAIR001,005,0*3E
```

\$PAIR006 - Cold start**Format**

```
$PAIR006*3C<CR><LF>
```

Return format

```
$PAIR001,006,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send $PAIR006*3C // Receiver response $PAIR001,006,0*3D
```

\$PAIR023 - System reboot**Format**

```
$PAIR023*3B<CR><LF>
```

Return format

None. The module reboots immediately without sending a response.

Example

```
// Host send $PAIR023*3B
```

\$PAIR513 - Save settings to NVRAM**Format**

```
$PAIR513*3D<CR><LF>
```

Return format

```
$PAIR001,513,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send $PAIR513*3D // Receiver response $PAIR001,513,0*3C
```

\$PAIR514 - Restore factory defaults**Format**

```
$PAIR514*3A<CR><LF>
```

Return format

```
$PAIR001,514,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send $PAIR514*3A // Receiver response $PAIR001,514,0*3B
```

\$PAIR066 - Set GNSS search mode**Format**

```
$PAIR066,<GPS>,<GL0>,<GAL>,<BDS>,<QZSS>,<NavIC>*<Checksum><CR><LF>
```

Parameters

Field	Description
Constellation flags	0 : Disable 1 : Enable

Return format

```
$PAIR001,066,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send: enable GPS, GLONASS, Galileo, BeiDou, and QZSS; disable NavIC $PAIR066,1,1,1,1,1,0*3B // Receiver response $PAIR001,066,1*3A
```

\$PAIR067 - Get GNSS search mode**Format**

```
$PAIR067*3B<CR><LF>
```

Return format

```
$PAIR067,<GPS>,<GL0>,<GAL>,<BDS>,<QZSS>,<NavIC>*<Checksum><CR><LF>
```

Return parameters

Field	Description
Constellation flags	0 : Disabled 1 : Enabled

Example

```
// Host send $PAIR067*3B // Receiver response $PAIR001,067,0*3A $PAIR067,1,0,0,0,0,0*3A
```

\$PAIR050 - Set position fix interval

GUIDE K35C is specified for a 1 Hz navigation update rate. GUIDE K35E is the RTK + DR variant and supports update intervals down to 100 ms , corresponding to 10 Hz output.

Format

```
$PAIR050,<Interval_ms>*<Checksum><CR><LF>
```

Parameters

Field	Description
Interval_ms	Position fix interval in milliseconds. Valid range: 100 to 1000 . 1000 : 1 Hz, default 200 : 5 Hz 100 : 10 Hz

Return format

```
$PAIR001,050,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send: set 10 Hz $PAIR050,100*22 // Receiver response $PAIR001,050,0*3E
```

\$PAIR051 - Get position fix interval**Format**

```
$PAIR051*3E<CR><LF>
```

Return format

```
$PAIR051,<Interval_ms>*<Checksum><CR><LF>
```

Return parameters

Field	Description
Interval_ms	Position fix interval in milliseconds.

Example

```
// Host send $PAIR051*3E // Receiver response $PAIR001,051,0*3F $PAIR051,1000*13
```

\$PAIR031 - Get 3D velocity**Format**

```
$PAIR031*38<CR><LF>
```

Return format

```
$PAIR031,<VX>,<VY>,<VZ>,<Acc>*<Checksum><CR><LF>
```

Return parameters

Field	Description
VX	WGS84 ECEF X Cartesian velocity vector in m/s.
VY	WGS84 ECEF Y Cartesian velocity vector in m/s.
VZ	WGS84 ECEF Z Cartesian velocity vector in m/s.
Acc	Three-dimensional speed 1-sigma accuracy in m/s.

Example

```
// Host send $PAIR031*38 // Receiver response $PAIR001,031,0*39 $PAIR031,0.19,-0.07,-0.11,0.49*3A
```

\$PAIR864 - Set baud rate

The baud-rate change takes effect immediately after the next response. Update the host UART configuration before sending the next command. To keep the change after a power cycle, send \$PAIR513 to save the settings, then reboot with \$PAIR023 or cycle the power supply.

Format

```
$PAIR864,<Port_Type>,<Port_Index>,<Baudrate>*<Checksum><CR><LF>
```

Parameters

Field	Description
Port_Type	0 : UART
Port_Index	0 : UART0, main communication port
Baudrate	9600 , 115200 default, 230400 , 460800 , or 921600 .

Return format

```
$PAIR001,864,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send: set UART0 baud rate to 115200 $PAIR864,0,0,115200*1B // Receiver response: sent at the current baud rate
```

\$PAIR865 - Get baud rate**Format**

```
$PAIR865,<Port_Type>,<Port_Index>*<Checksum><CR><LF>
```

Parameters

Field	Description
Port_Type	0 : UART
Port_Index	0 : UART0, main communication port

Return format

```
$PAIR865,<Port_Type>,<Port_Index>,<Baudrate>*<Checksum><CR><LF>
```

Return parameters

Field	Description
Baudrate	Current UART baud rate.

Example

```
// Host send $PAIR865,0,0*31 // Receiver response $PAIR001,865,0*30 $PAIR865,0,0,115200*1A
```

\$PAIR062 - Set NMEA output rate**Format**

```
$PAIR062,<Type>,<Output_Rate>*<Checksum><CR><LF>
```

Parameters

Field	Description
Type	-1 : Reset all sentences to default 0 : NMEA_SENT_GGA 1 : NMEA_SENT_GLL 2 : NMEA_SENT_GSA 3 : NMEA_SENT_GSV 4 : NMEA_SENT_RMC 5 : NMEA_SENT_VTG 6 : NMEA_SENT_ZDA 7 : NMEA_SENT_GRS 8 : NMEA_SENT_GST
Output_Rate	0 : Disabled; do not output the sentence. 1 : Output once every position fix. 2 : Output once every two position fixes. Valid range: 0 to 20 .

Return format

```
$PAIR001,062,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send: disable GSV output $PAIR062,3,0*3D // Receiver response $PAIR001,062,0*3F
```

\$PAIR063 - Get NMEA output rate**Format**

```
$PAIR063,<Type>*<Checksum><CR><LF>
```

Parameters

Field	Description
Type	NMEA sentence type. Use -1 to return all configurations or 0 to 8 for a selected sentence.

Return format

```
$PAIR063,<Type>,<Output_Rate>*<Checksum><CR><LF>
```

Return parameters

Field	Description
Output_Rate	Current output interval setting from 0 to 20 .

Example

```
// Host send $PAIR063,0*23 // Receiver response $PAIR001,063,0*3E $PAIR063,0,3*3C
```

\$PAIR098 - Set NMEA decimal precision**Format**

```
$PAIR098,<MODE>*<Checksum><CR><LF>
```

Parameters

Field	Description
MODE	0 : Latitude and longitude with four digits; altitude with one digit. 1 : Latitude and longitude with five digits; altitude with two digits. 2 : Latitude and longitude with six digits; altitude with three digits. 3 : Latitude and longitude with seven digits; altitude with three digits.

Return format

```
$PAIR001,098,<Result>*<Checksum><CR><LF>
```

Example

```
// Host send $PAIR098,0*27 // Receiver response $PAIR001,098,0*3A
```

\$PAIR099 - Get NMEA decimal precision**Format**

```
$PAIR099*3A<CR><LF>
```

Return format

```
$PAIR099,<MODE>*<Checksum><CR><LF>
```

Return parameters

Field	Description
MODE	Current precision mode from 0 to 3 .

Example

```
// Host send $PAIR099*3A // Receiver response $PAIR001,099,0*3B $PAIR099,0*26
```

\$PAIR100 - Set NMEA output mode**Format**

```
$PAIR100,<NMEA_MODE>,<PROPRIETARY_MODE>*<Checksum><CR><LF>
```

Parameters

Field	Description
NMEA_MODE	0 : Disable NMEA. 1 : ASCII NMEA v4.1, default. 2 : ASCII NMEA v3.0.
PROPRIETARY_MODE	0 : Disable additional proprietary sentences, default. 1 : Enable proprietary sentences.

Return format

\$PAIR001,100,<Result>*<Checksum><CR><LF>

Example

// Host send: ASCII NMEA v4.1, additional proprietary sentence disabled \$PAIR100,1,0*3A // Receiver response \$PAIR001,100,0*3A

\$PAIR101 - Get NMEA output mode**Format**

\$PAIR101*3A<CR><LF>

Return format

\$PAIR101,<NMEA_MODE>,<PROPRIETARY_MODE>*<Checksum><CR><LF>

Return parameters

Field	Description
NMEA_MODE	Current standard NMEA output mode from 0 to 2 .
PROPRIETARY_MODE	Current proprietary sentence output state from 0 to 1 .

Example

// Host send \$PAIR101*3A // Receiver response \$PAIR001,101,0*3B \$PAIR101,0,1*3B

Integration references

Continue with the GUIDE specifications, carrier-board interface definitions, or the hardware integration overview.